Linear models

1

a $0.5x + 1.5y = 5$

b 4 sheets

2

a $\$63.50$

b $\$5.50$

c A fixed cost such as a toll charge.

3

a

Time	0	1	2	3	4
Vertical velocity	120	88	56	24	-8

b -32

c The baseball is moving downwards instead of upwards at $T = 4$.

4

a $y = 45x + 150$

b 45

c The total amount charged increases by $\$45$ for each additional hour of work.

d 150

e The callout fee or the minimum amount charged by the carpenter.

f $\$420$

5

a $y = 75x + 50$

b 75

c Mohamad is travelling a distance of 75 mi every hour.

d $y = 50$

e The distance Mohamad had previously driven in his Subaru.

f 218.75 mi

6

a $y = 6.8x + 0.8$

b 6.8

c The cost of each additional minute.

d 0.8

e The initial connection fee.

f $\$41.60$

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7

a $y = 100 - 9x$

b -9

c Mario is running at a speed of $9\ \text{km/h}$.

d 100

e The length of the ultramarathon or the total distance Mario has to run.

f $59.5\ \text{km}$

8

a $y = 16x - 1300$

b 16

c The average amount the restaurant receives from each customer.

d -1300

e The fixed weekly cost

f $\$3660$

9

a $y = 10x + 340$

b 10

c His earnings per phone.

d 340

e The minimum amount earned in a week, or the fixed earnings per week independent of how many cellphones are sold.

f $\$700$

10

a 0.2 seconds **b** $y = -0.2x + 6$ **c** 5.1 seconds **d** $x = 2.5$ hours

11

a $1.4\ \text{yd/min}$ **b** $y = 1.4x$ **c** $8.4\ \text{yd}$ **d** 9 minutes

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12

a

Number of minutes passed (x)	0	2	4	6	8	10	12
Heart rate (y)	49	55	61	67	73	79	85

b 3

c $y = 3x + 49$

d The change in one minute of Mae's heartbeat.

13

a -0.51

b $y = -0.51x + 40$

c The cost of each text message.

d 40

e The amount of credit Paul purchased.

f $\$21.64$

14

a

Number of minutes passed (x)	0	5	10	15	20
Amount of fuel left in the tank (y)	150	125	100	75	50

b $y = 150 - 5x$

c The amount of fuel in the car is decreasing at a constant rate of 5 L per minute.

d 30 minutes

15

a $y = 25x + 85$

b 25

c The cost of manufacturing each chair.

d 85

e The fixed cost of manufacturing.

f $\$410$

16

a $y = -0.6x + 28$

b -0.6

c The rate at which the water level is falling per minute.

d 28

e The water level before any water is pumped out.

f 19 m

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Graphs of linear models

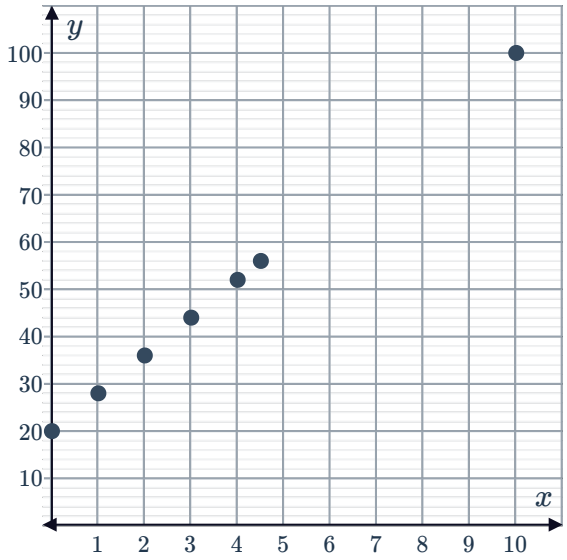
17

a

Number of hours passed (x)	0	1	2	3	4	4.5	10
Amount of water in tank (y)	20	28	36	44	52	56	100

b $y = 8x + 20$

c



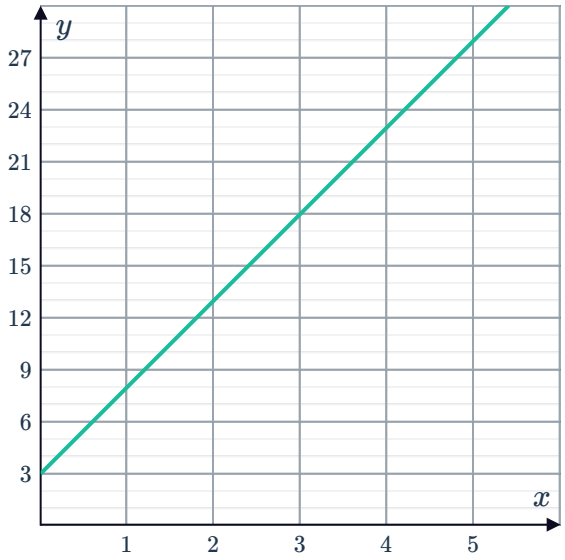
18

a 5°F

b 3°F

c $y = 5x + 3$

d



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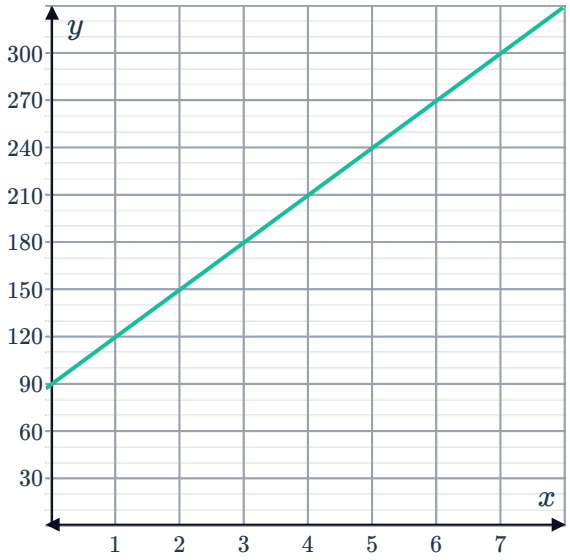
19

a $y = 30x + 90$

b

Number of hours worked (x)	0	1	2	3	4	4.5	7
Total fee charged (y)	90	120	150	180	210	225	300

c



20

a $\frac{1}{2}$

b 32

c $y = \frac{1}{2}t + 32$

d The amount by which the temperature is rising every minute.

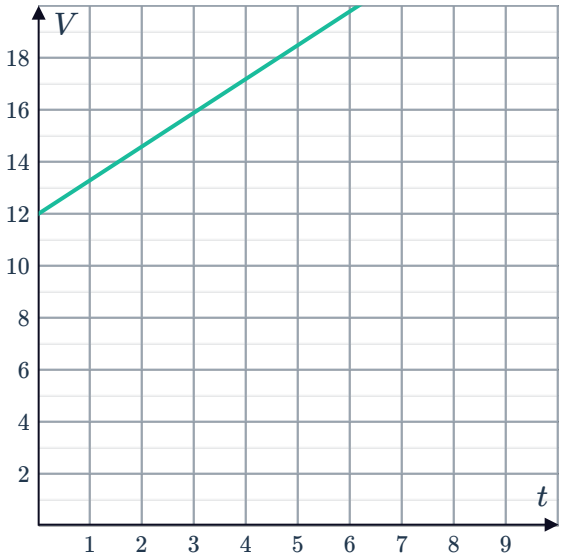
e The temperature of the room before the heater has been turned on.

f 52°F

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21

a



b $\frac{13}{10}$

c The rate of acceleration - that is, the rate of increase in the velocity.

d 12

e The initial speed of the ball.

f $V = 12 + \frac{13}{10}t$

g $36.7\ \text{m/s}$

22

a

Water temperature $(^\circ\text{F})$	-3	-2	-1	0	1	2	3
Surface air temperature $(^\circ\text{F})$	-7	-5	-3	-1	1	3	5

b $y = 2x - 1$

c 27°F

d 12°F

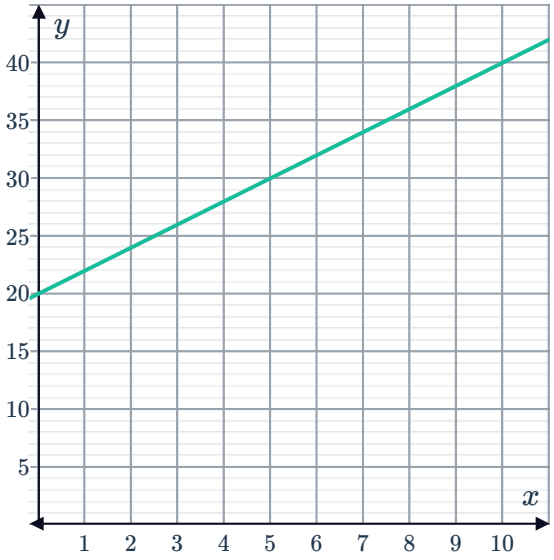
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23

a

Number of hours passed since midnight (x)	0	1	2	3	4	4.5	10
Amount of water in tank (y)	20	22	24	26	28	29	40

b



c $y = 2x + 20$

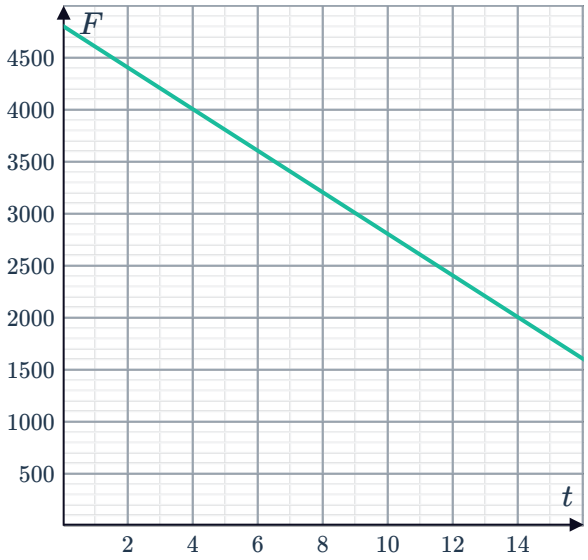
d 20

e 2 pm

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24

a



b -200

c The rate of change of the fish population each year.

d $F = 4800$

e $F = -200t + 4800$

f 2200 fish

g $t = 14$

25

a $-\frac{1}{2}$

b 30

c $y = -\frac{1}{2}x + 30$

d The amount of water that is flowing out of the hole every minute.

e The capacity of the bucket.

f $3\ \text{L}$

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